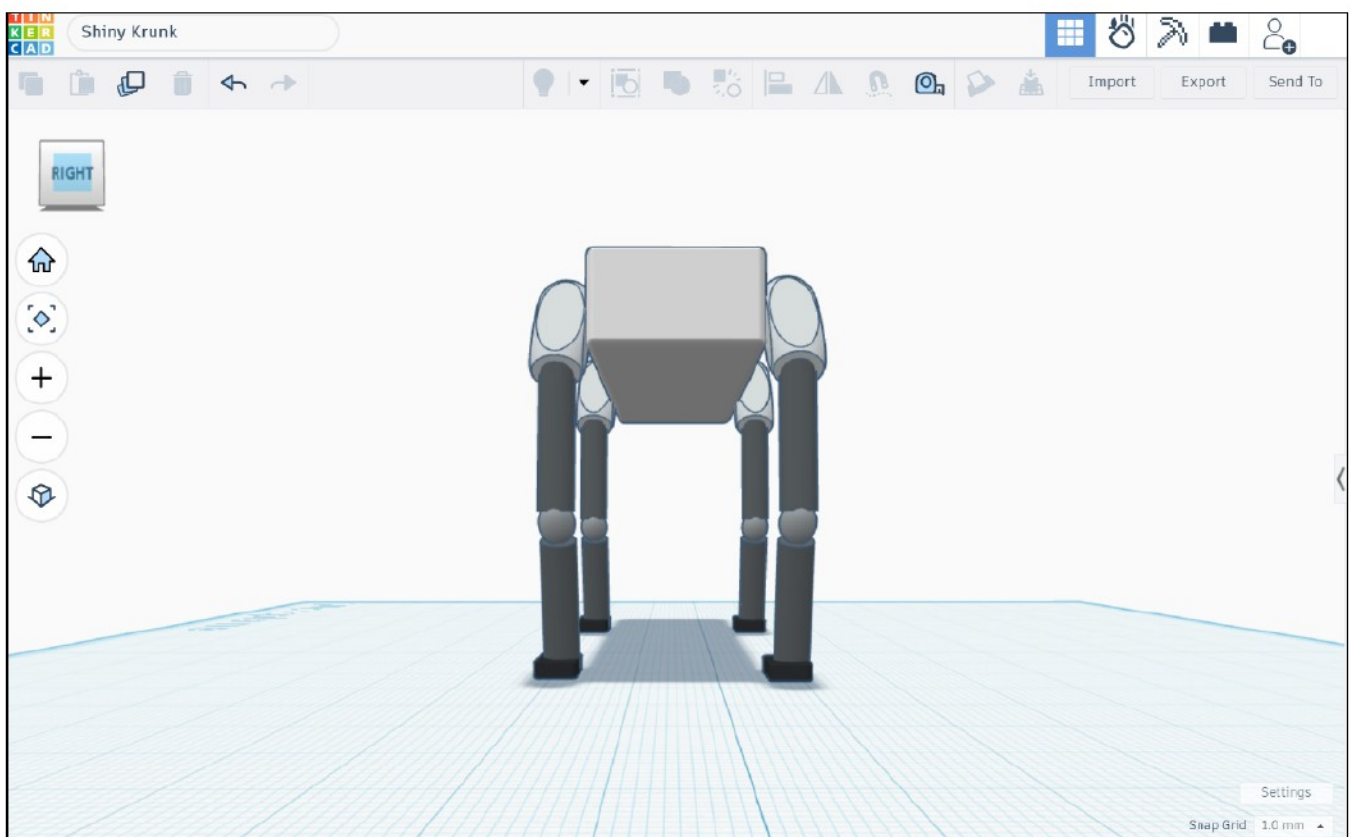
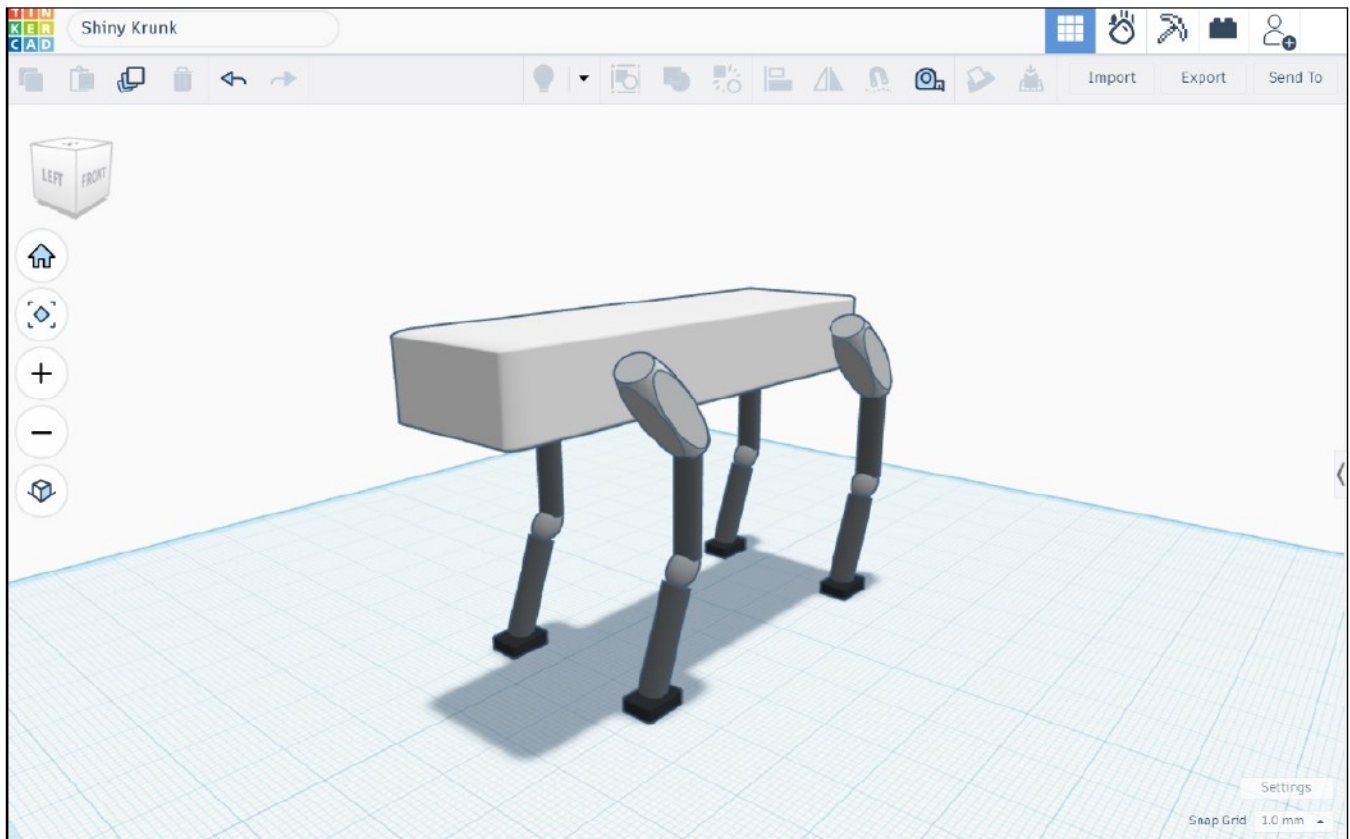
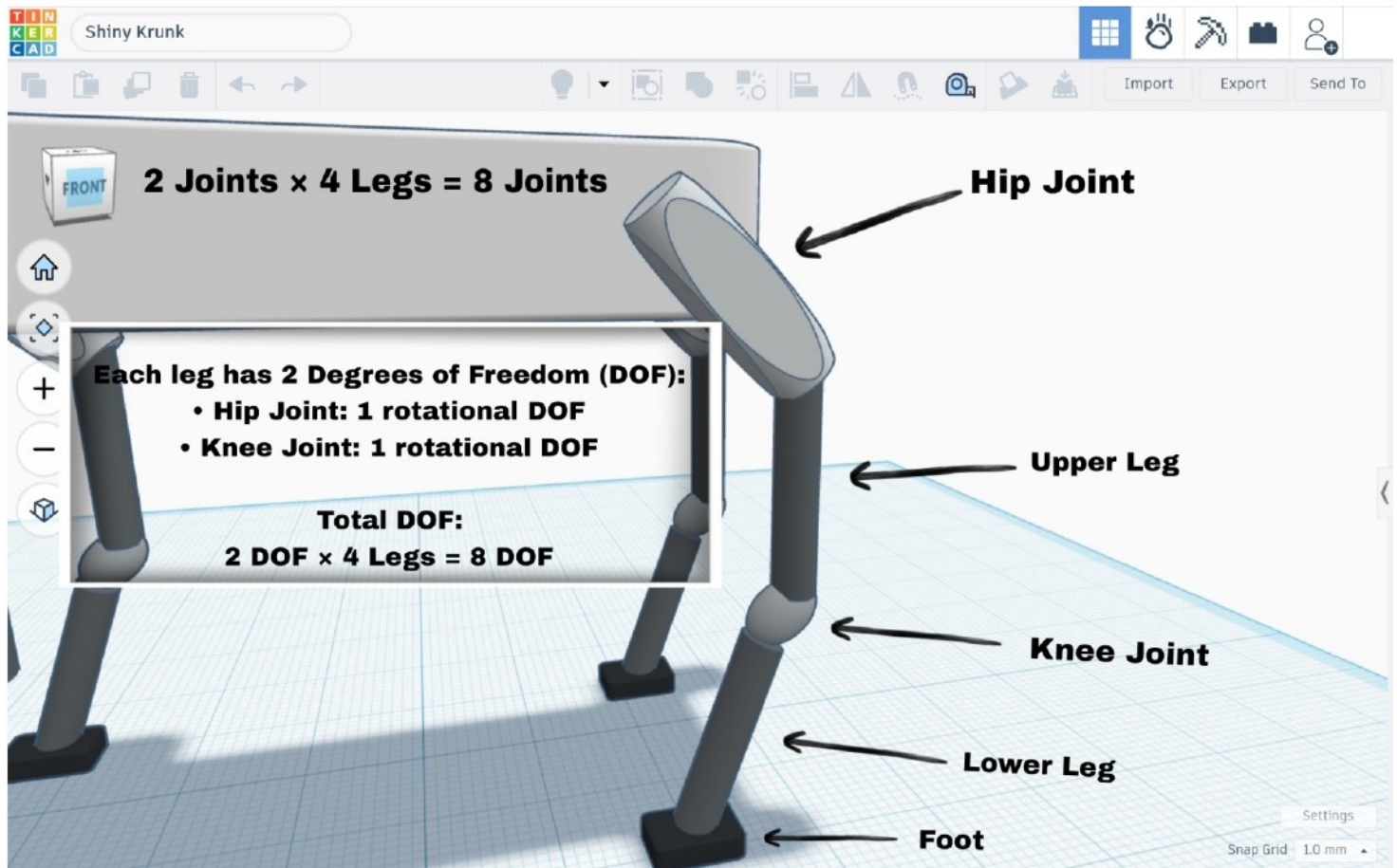


Preliminary Design of a Robotic Dog

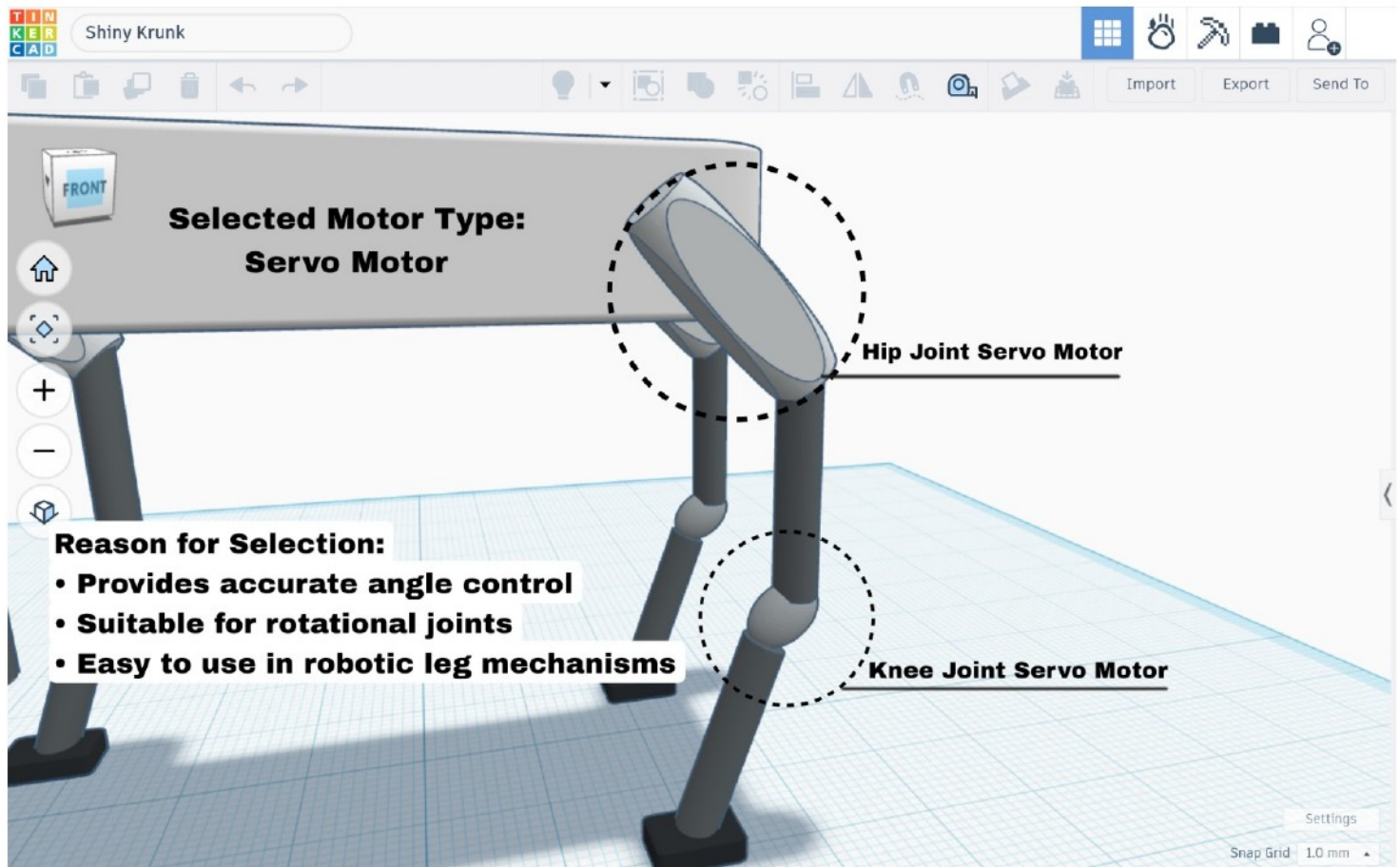
Created by : Rayan Hamdan Alzaedi



Joints and Degrees of Freedom



Motor Selection



Preliminary Torque Calculation

TIN KEN CAD Shiny Krunk

Assumptions:

- Robot mass = 2 kg
- Number of legs = 4
- Load per leg = $2 \div 4 = 0.5$ kg
- Gravity = 9.81 m/s^2
- Lever arm distance = 0.10 m

Force:

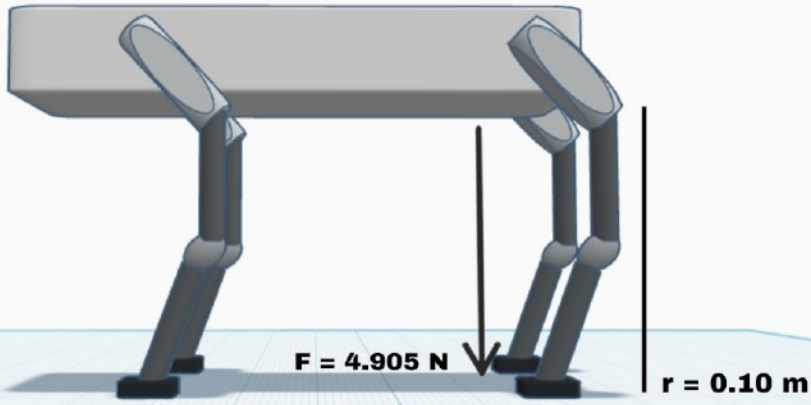
$$F = m \times g$$
$$F = 0.5 \times 9.81$$
$$F = 4.905 \text{ N}$$

Torque:

$$\tau = F \times r$$
$$\tau = 4.905 \times 0.10$$
$$\tau = 0.4905 \text{ N}\cdot\text{m}$$

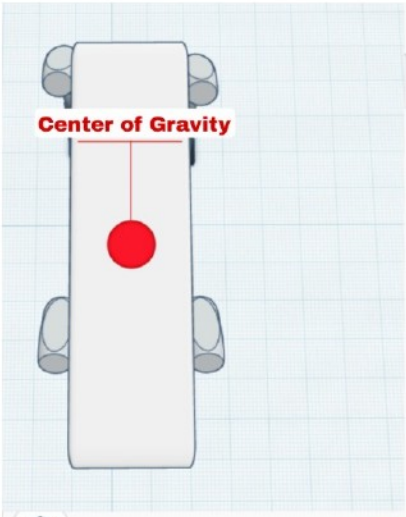
Initial required torque $\approx 0.49 \text{ N}\cdot\text{m}$

Recommended motor torque:
At least 1 N·m



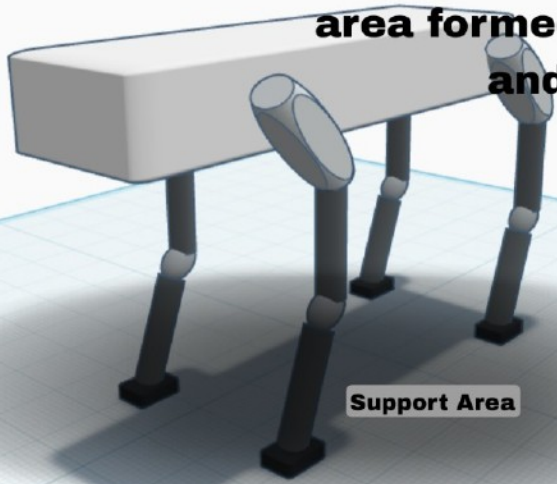
Settings
Snap Grid 1.0 mm

Stability and Center of Gravity



Center of Gravity

The stability of the robotic dog depends on keeping the center of gravity inside the support area formed by the four legs and flat feet.



Support Area

Settings
Snap Grid 1.0 mm

Proposed Gait

The proposed walking method for this robotic dog is the Crawl Gait.

In this gait, one leg moves at a time while the other three legs remain on the ground to maintain stability.

Suggested sequence:

- 1. Front Right Leg**
- 2. Rear Left Leg**
- 3. Front Left Leg**
- 4. Rear Right Leg**

This gait is simple, stable, and suitable for a preliminary robotic dog design.

Expected Mechanical Problems

Possible mechanical problems in this design include:

- Insufficient motor torque**
- Joint friction**
- Unbalanced weight distribution**
- Leg slipping on smooth surfaces**
- Weak body-leg connections**
- Limited joint movement range**